

# AGRICULTURAL SCIENCE - LESSON 01

## Contributing Effectively to Agricultural Development in Sri Lanka

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### 1. Economic Overview & Role of Agriculture

#### Gross Domestic Product (GDP)

**Gross Domestic Product (GDP)** is the total value of all final goods and services produced within a country during a specific year.

#### Sectoral Contribution Ranking to GDP:

Services Sector > Industrial Sector > Agricultural Sector

#### Gross National Product (GNP)

**Gross National Product (GNP)** is the total value of final goods and services produced by the citizens of a country in a year (excludes foreign investment within the country, but includes earnings sent home by citizens working abroad).

#### Key Agricultural Sub-sectors Contributing to GNP:

- **Crop Production:** Paddy (Rice), Tea, Rubber, Coconut
- **Livestock Production:** Milk, Eggs, Meat
- **Fisheries Production:** Inland Fresh Water Fisheries, Marine Fisheries
- **Forestry Resources:** Forest cover accounts for up to 25% of the land area

#### Employment in the Agricultural Sector

**Sectoral Employment Contribution Ranking:** Services > Industry > Agriculture

*2014 Statistical Breakdown:*

- **Agriculture:** 10.1%
- **Industry:** 32.3%
- **Services:** 57.6%

### 2. Agriculture-Related Industries & Services

#### A. Agriculture-Related Industries (Agro-Industries)

- **Food Processing:** Rice processing, biscuits, bread, yogurt

- **Agricultural Inputs Production:** Fertilizers, machinery, agrochemicals
- **Agricultural Infrastructure & Construction:** Farm buildings, animal housing, reservoirs & irrigation channels
- **Animal Feed Production:** Broiler feeds, layer poultry feeds

### B. Agriculture-Related Services

Agricultural Extension Services, Agricultural Research, Transport Services, Maintenance Services, Agricultural Credit & Insurance Services, Marketing of Agricultural Goods, and Provision of Trained Labor.

## 3. Historical Self-Sufficiency & Modern Challenges

### Historical Evidence of a Self-Sufficient Economy

Historical records prove Sri Lanka was self-sufficient, evidenced by rice exports during the reign of King Parakramabahu, an advanced indigenous medical system, and maintaining sufficient food supplies during famines.

#### Key Drivers of Historical Agricultural Prosperity:

- Advanced Irrigation Engineering & Technology
- State Support & Royal Patronage
- Cultural and Religious Alignment
- Peasant Settlements & Multipurpose Schemes (e.g., Gal Oya [First], Mahaweli [Largest], Uda Walawe, Inginiyitiya)
- **Green Revolution:** Enhanced unit land yield through high-yielding varieties, mechanization, and intensive input utilization (brought both positive and negative impacts).

### Modern Agricultural Challenges & Mitigative Solutions

| Agricultural Challenge                                 | Mitigative Solution / Strategy                                      |
|--------------------------------------------------------|---------------------------------------------------------------------|
| Binding conditions/agreements during food importations | Reduce reliance on imports by expanding domestic food production    |
| Outdated/Backward Technology                           | Introduce and adopt modern agricultural technologies                |
| Failure to reach maximum yield potential               | Utilize improved crop varieties to reach peak yield potential       |
| Scarcity of agri-entrepreneurship                      | Encourage existing entrepreneurs and inspire new agri-entrepreneurs |
| Adverse climatic impacts                               | Cultivate stress-tolerant & climate-resilient crops                 |
| Depletion of genetic resources (Gene pool)             | Conserve and improve indigenous gene pools                          |

## 4. Future Trends, Policies & Legislation

### Future Trends in Agriculture

- Maximizing land utilization efficiency.
- Enhancing the qualitative standards of the agricultural workforce.
- Striving for optimal control over climatic factors.
- Establishing national food security.

### National Agricultural Policy & Acts

A National Agricultural Policy comprises standard principles established to advance agriculture. Key policies cover irrigation, plant quarantine, production/trade/consumption, land distribution, and chemical usage.

#### Acts Passed in Parliament:

- **Agrarian Development Act No. 46 of 2000:** Protected rights of tenant farmers and prevented illegal conversion of paddy lands to other uses.
- **Seed Certification Act No. 23 of 2003**
- **Water Management Act | Plant Protection Act | Environmental Act | Soil Conservation Act | Quarantine Act | Control of Pesticides Act**

## 5. Key Institutions Supporting Agricultural Development

### State Research Institutions

| Crop Sector         | Primary Research Centers / Stations |
|---------------------|-------------------------------------|
| Horticultural Crops | Gannoruwa                           |
| Field Crops         | Mahailluppallama                    |
| Paddy (Rice)        | Batalagoda, Labuduwa, Ambalantota   |
| Rubber              | Agalawatta                          |
| Tea                 | Talawakele                          |
| Cinnamon            | Kamburupitiya                       |
| Sugarcane           | Uda Walawe                          |
| Coconut             | Lunuwila                            |

### Institutional Network Categories

- **Extension Services:** Department of Agriculture, Mahaweli Authority

- **Agricultural Inputs:** Department of Irrigation (Water supply)
- **Marketing & Trade:** Livestock Development Board, Sri Lanka Tea Board
- **Private Enterprises:** CIC Holdings, Hayleys Agriculture, A. Baur & Co.
- **Non-Governmental Organizations (NGOs):** World Vision, SEEDO Sri Lanka, Sarvodaya
- **Financial & Credit Institutions:** Seylan Bank and Agrarian Banks (Agri-loans and subsidies)
- **Educational Institutions:** Universities (Peradeniya, Wayamba, Rajarata, Ruhuna, Uva Wellassa, Sabaragamuwa), Schools, Agricultural Colleges

### Key Roles of Key State Agencies

- **Department of Agriculture:** Conducts research, plans crop production, offers advisory services, diffuses new technology, conducts training programs, and enforces pesticide/soil conservation acts.
- **Department of Export Agriculture:** Conducts research on export crops and explores new market potentials.
- **Department of Agrarian Development:** Supplies inputs and services, grants credit via Agrarian Banks, solves farmer grievances, holds seasonal meetings, and forms farmer organizations.
- **Coconut Development Authority / Board:** Provides quality planting material and extension advice.